lab6

2026-02-25

Q1

a + b

```
## [1] 0.4444445 0.4444444 0.1111111 3.7000000 0.6000000 2.5000000 0.2549510
## [8] 0.3605551 0.0000000
```

```
## [1]    0.4444445    0.4444444    0.1111111 3699.9999753  600.0000017
## [6] 2500.0000000  254.9510311  360.5551318    0.0000000
```

We see that multiplying the data gives us the same parameters. They come in the order p1, p2, p3, mu1, mu2, mu3, sigma1, sigma2, sigma3.

Cosine similarity measures the cos of the angle between our current parameter vector and the previous one. It is independent of a scaling of the mu and sigma. But if we compare different types of parameters in the same cosine similarity, say means hovering around 1000 and fractions between 0-1 then the change in means dominate the cosine similarity. The example below shows this empirically, the cosine similarity changes more if the larger dimension changes by 80% and the smaller by 20% compared to the smaller changing by 80% and the larger dimension changing by 20%.

```
# this example shows
cossim(c(10, 1), c(12, 0.2))
```

```
##           [,1]
## [1,] 0.9965572
```

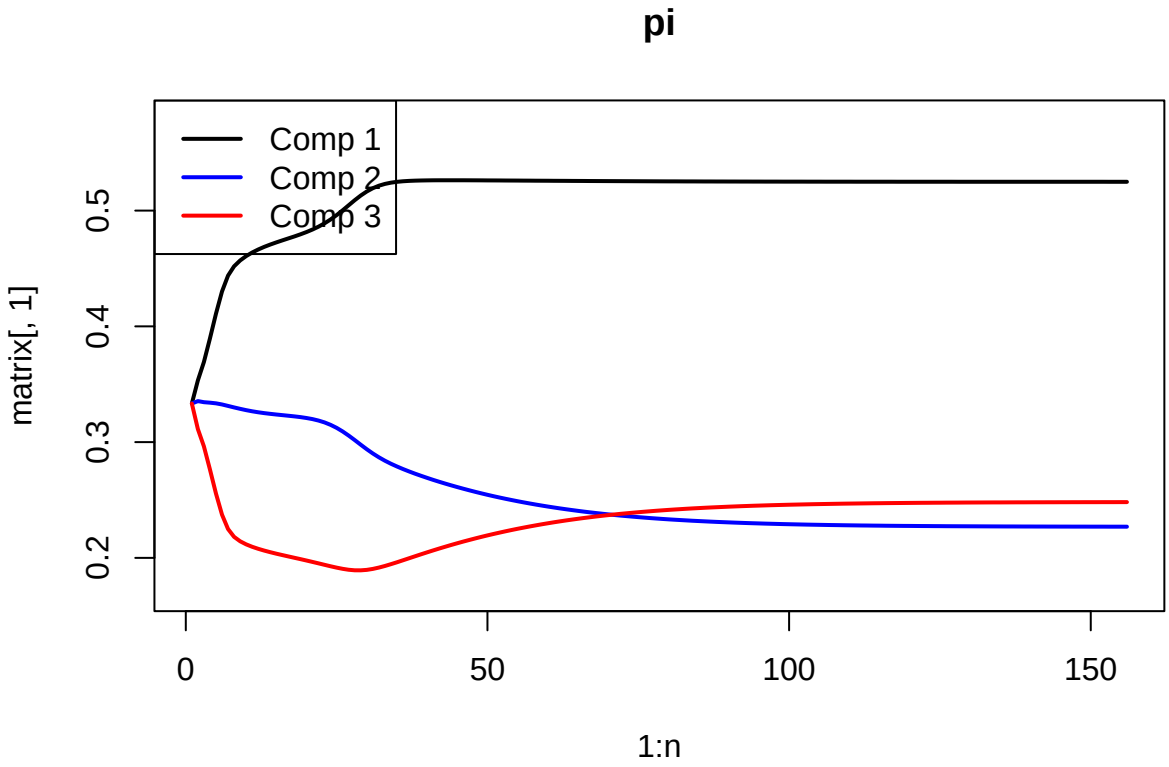
```
cossim(c(10, 1), c(2, 1.2))
```

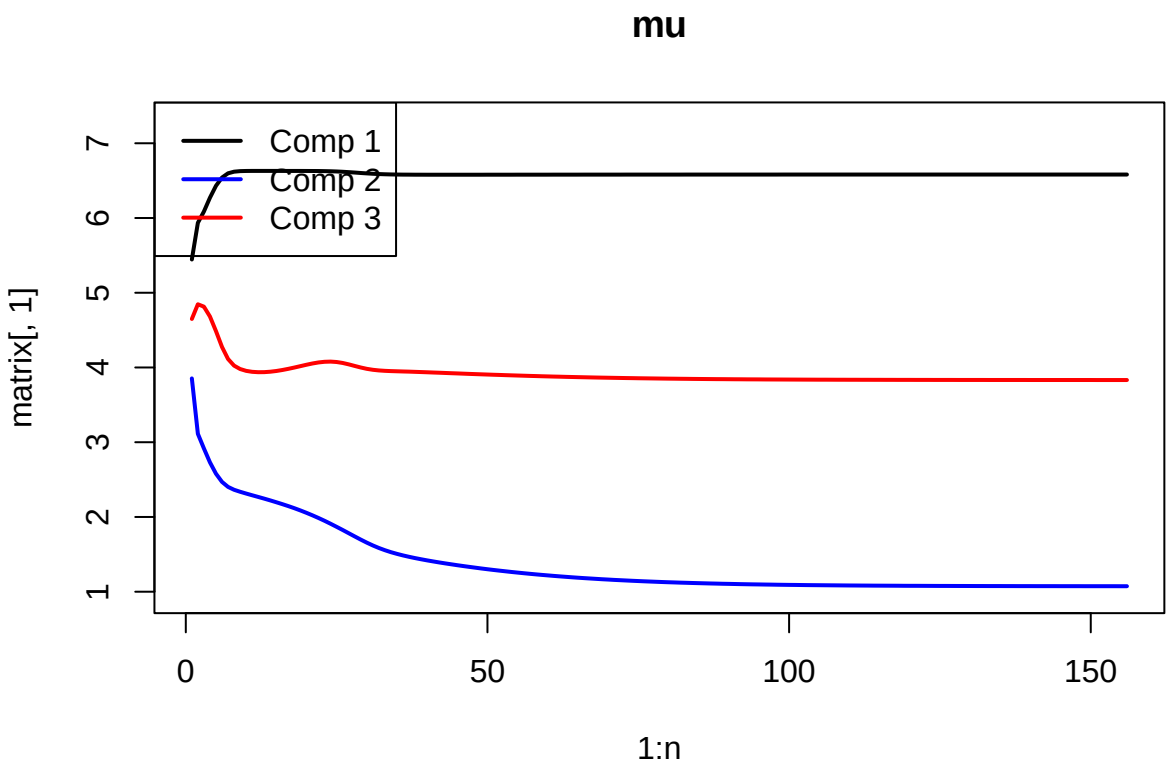
```
##           [,1]
## [1,] 0.9044316
```

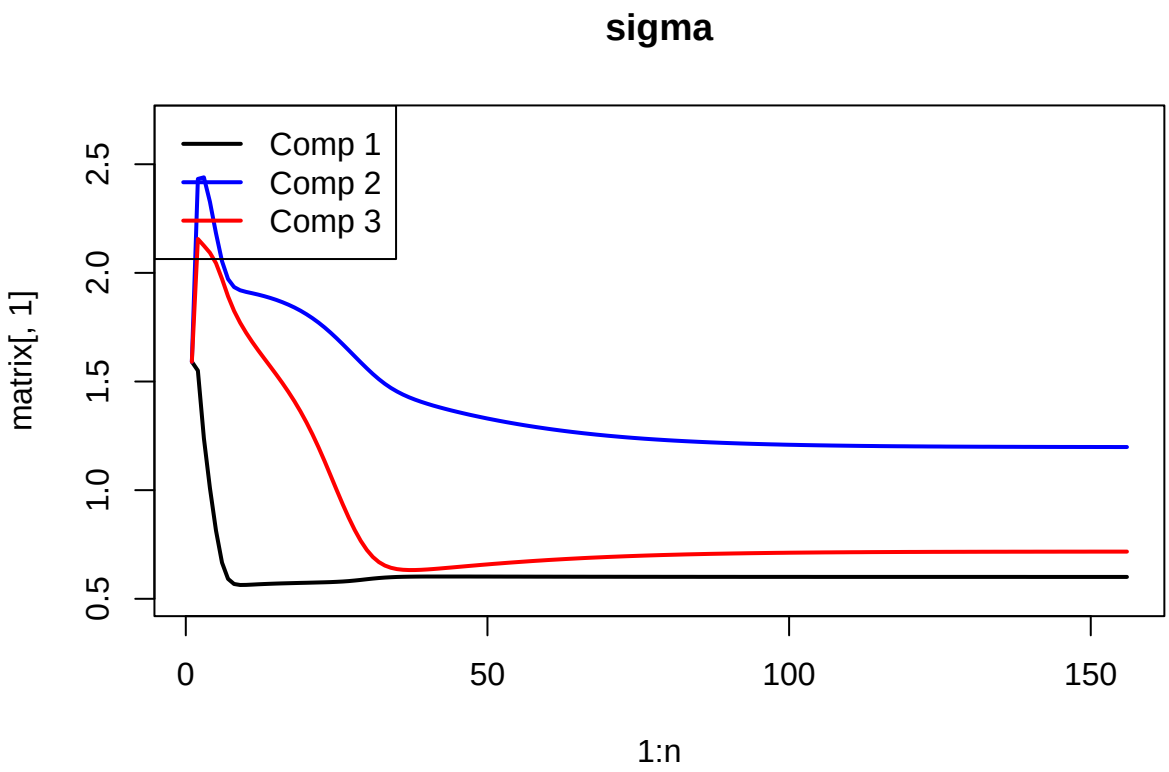
The parameters are on such different scales, using cosine similarity if mu1 changes by 10 it is going to overpower a change of 0.01 in p2. $cos(v1, v2)$ denotes the cosine of the angle between v1 and v2. 1 - cos as cos = 1 when the vecotors are perfectly alligned.

$$cc = \frac{(1 - cos(v_p, v1_p), (1 - cos(v_\mu, v1_\mu), (1 - cos(v_\sigma, v1_\sigma))}{3}$$

```
## [1] 0.5255415 0.2387176 0.2357409 6.5803106 1.1718721 3.8658191 0.6015409
## [8] 1.2558208 0.6901542
```

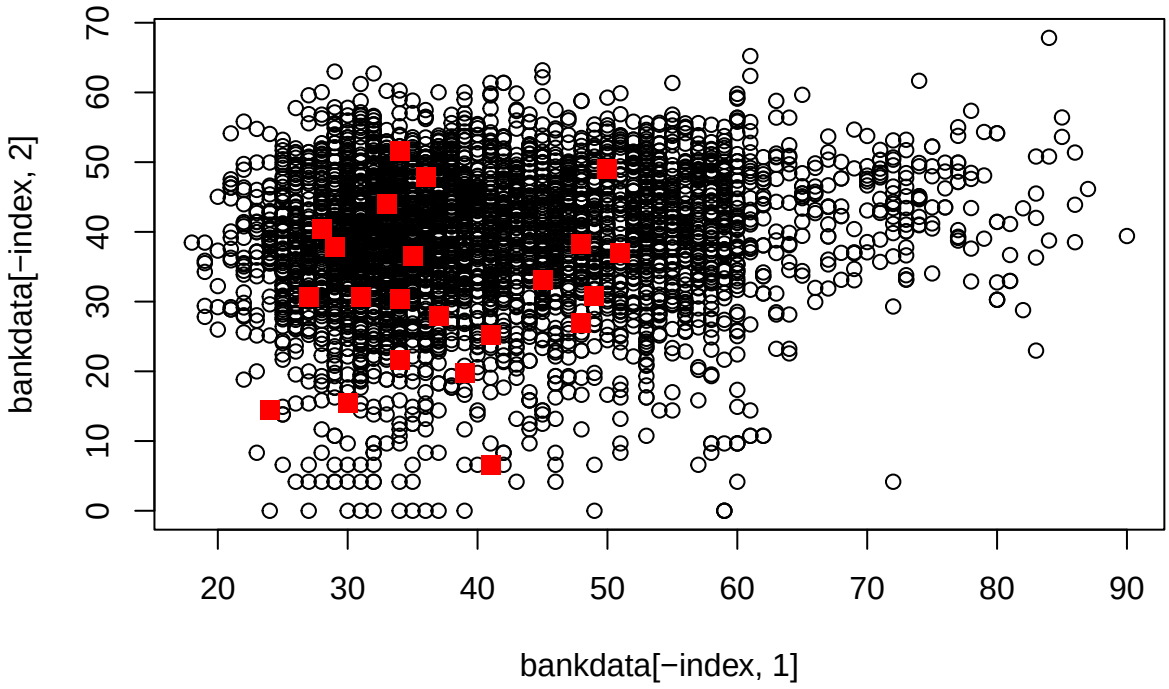

pi
Comp 1
Comp 2
Comp 3
matrix[, 1]
0.2
0.3
0.4
0.5
0
50
100
150
1:n


mu
Comp 1
Comp 2
Comp 3
matrix[, 1]
1
2
3
4
5
6
7
0
50
100
150
1:n



The plots support convergence as they all flatten out. The variance took the longest to converge.

Q2

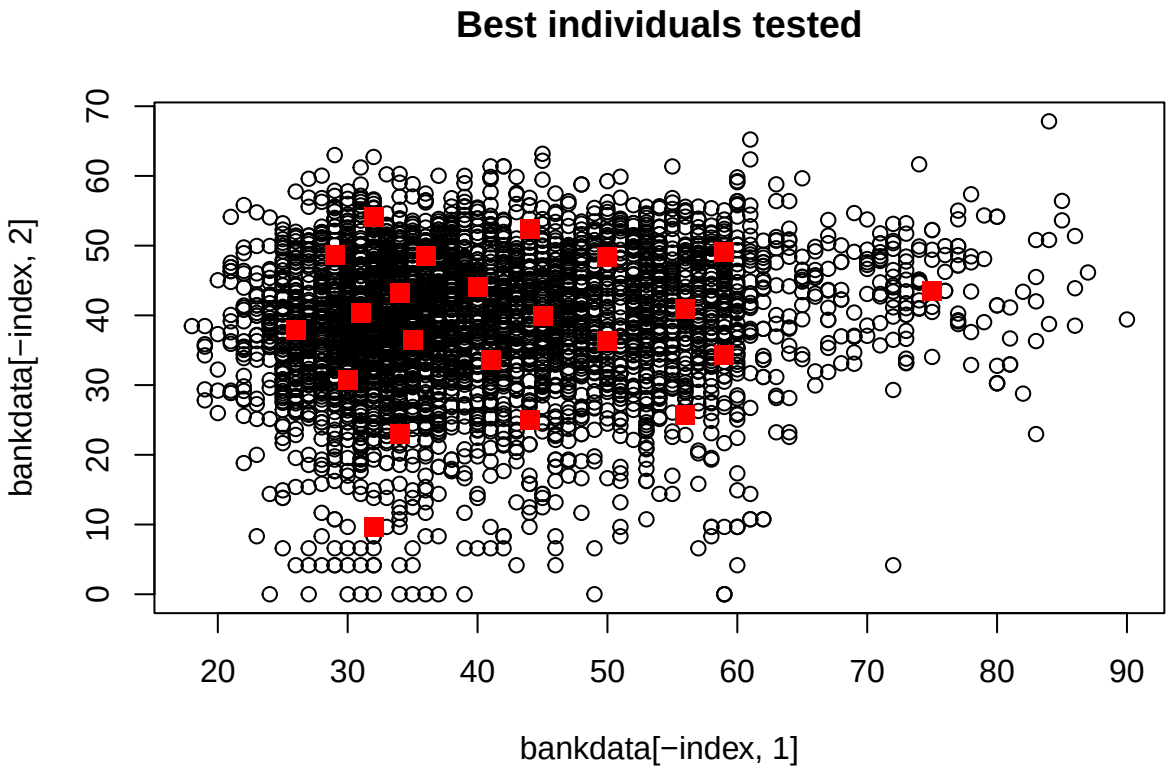


The proposal is a uniform mask, for each individual we swap them out with probability 1-prob_keep.

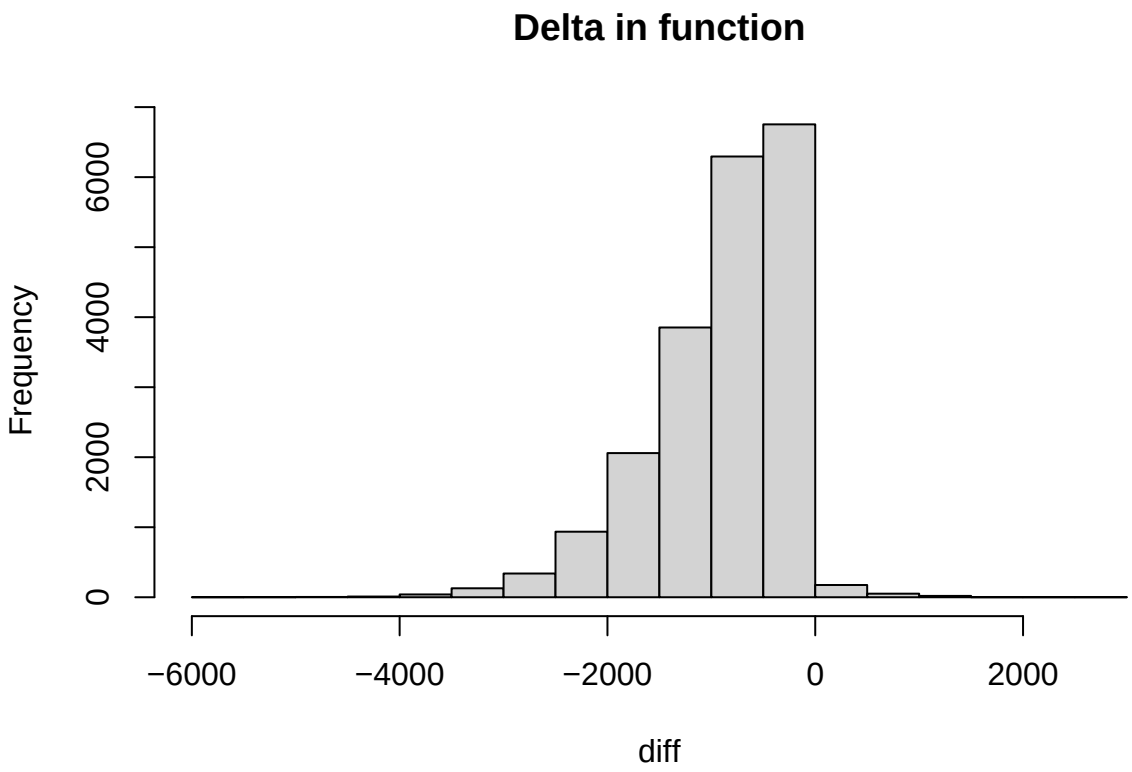
SA(prob_keep = 0.5, alpha = 0.9, beta = 1.02, mj = 50, temp = 10000, it = 200) This gives 16203 as minimum function value but is painfully slow so cannot run it in the knit. The convergence plot was flat after 100 itterations.

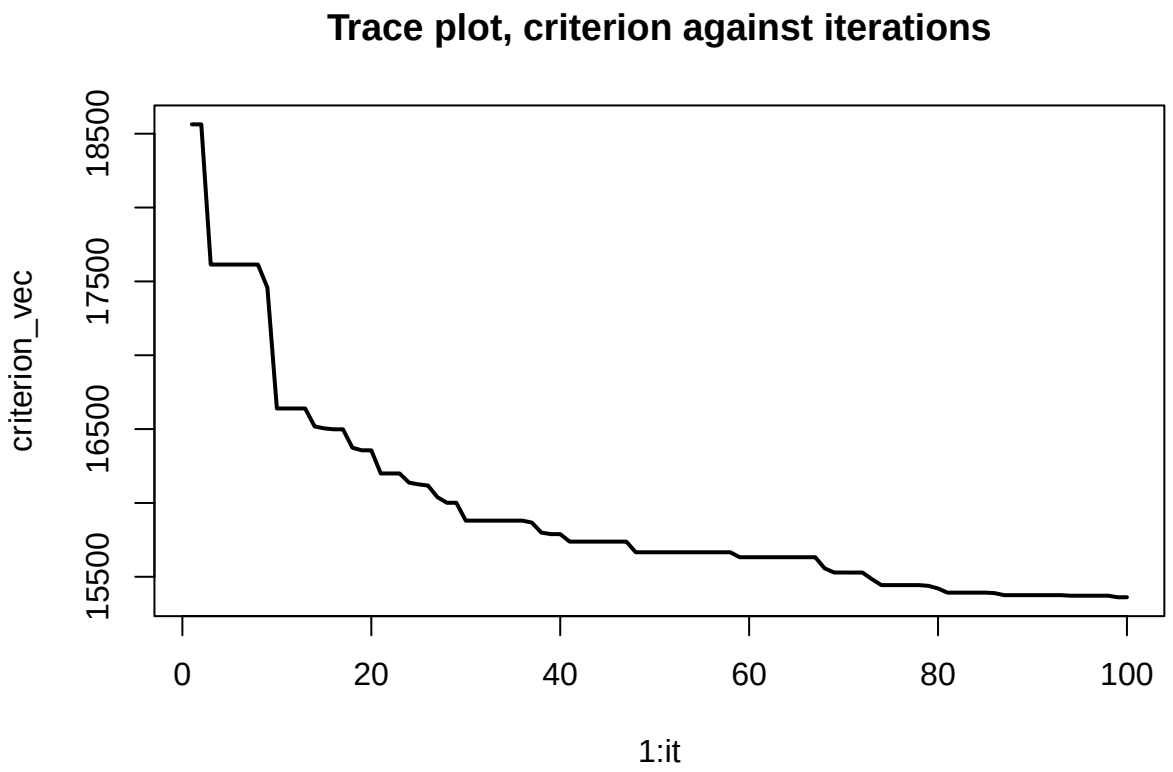
Lower function value and faster, higher starting temperature and lower alpha makes it decay faster, 0.9 probability swaps 2 people on average. Too many itterations with high alpha did not perform as well. The delta plot shows the average function difference and was used to find a good value for the temperature as the scale was unknown. The initial temperature was set such that most values would be accepted to be able to explore more at the start.

```
## [1] 20
## [1] 40
## [1] 60
## [1] 80
## [1] 100
## [1] "Starting function value:"
## [1] 21936.02
```




```
## [1] "Best function value:"
## [1] 15361.6
## [1] "Average diff:"
## [1] -870.7404
```

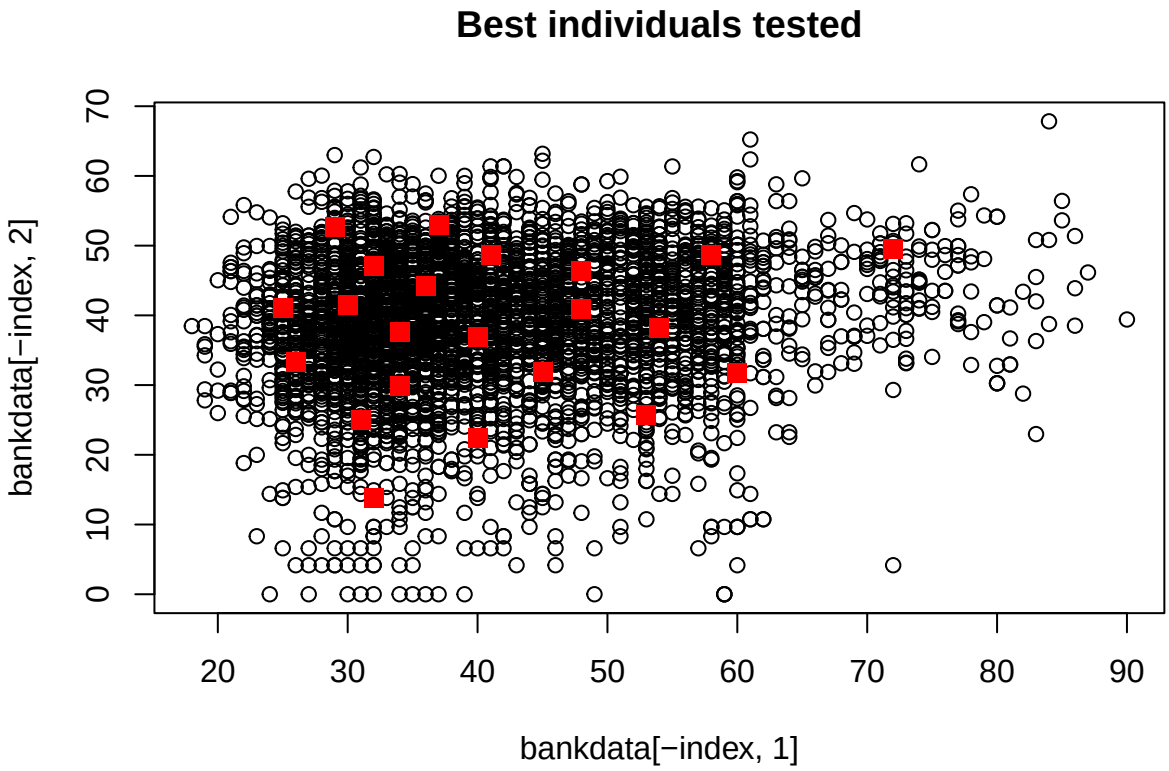

Delta in function
Frequency
0
2000
4000
6000
−6000
−4000
−2000
0
2000
diff



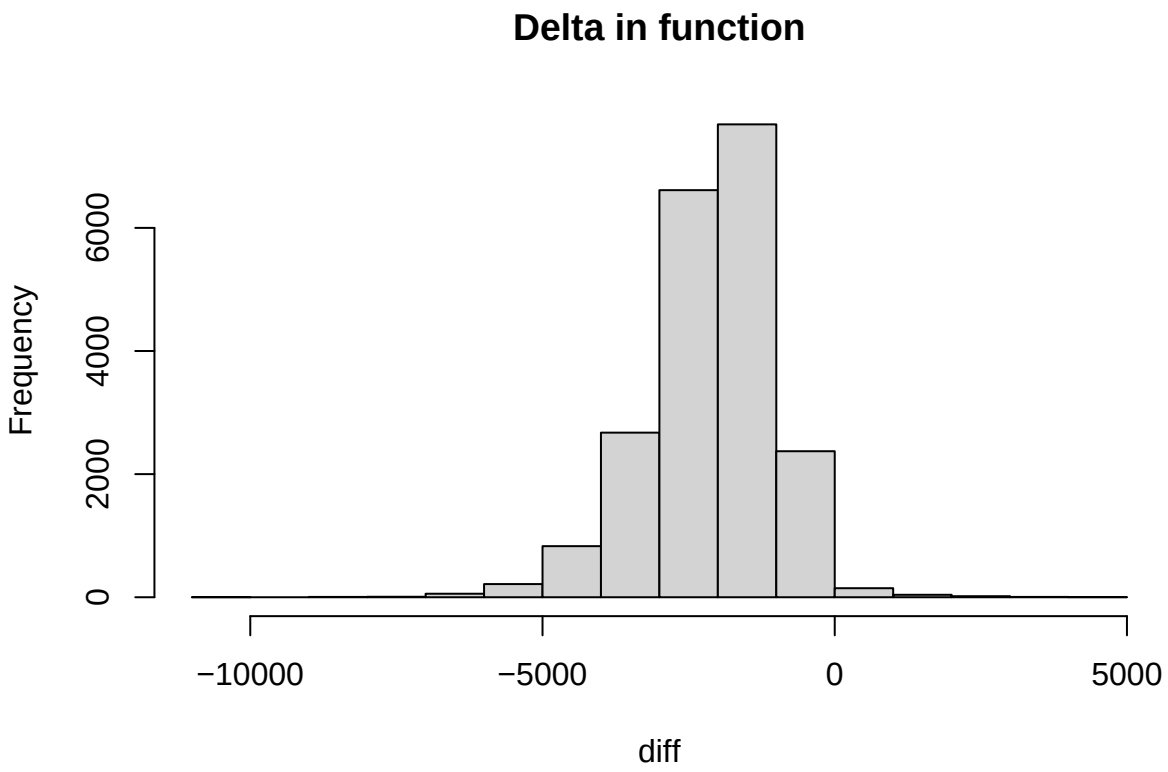

```
##  [1] 1747 1580  997    7  521  272 2462  212  614 3325 2041 1533  638  174 4116
## [16]  999 3702 3365 1454 3482 1556 1441
```

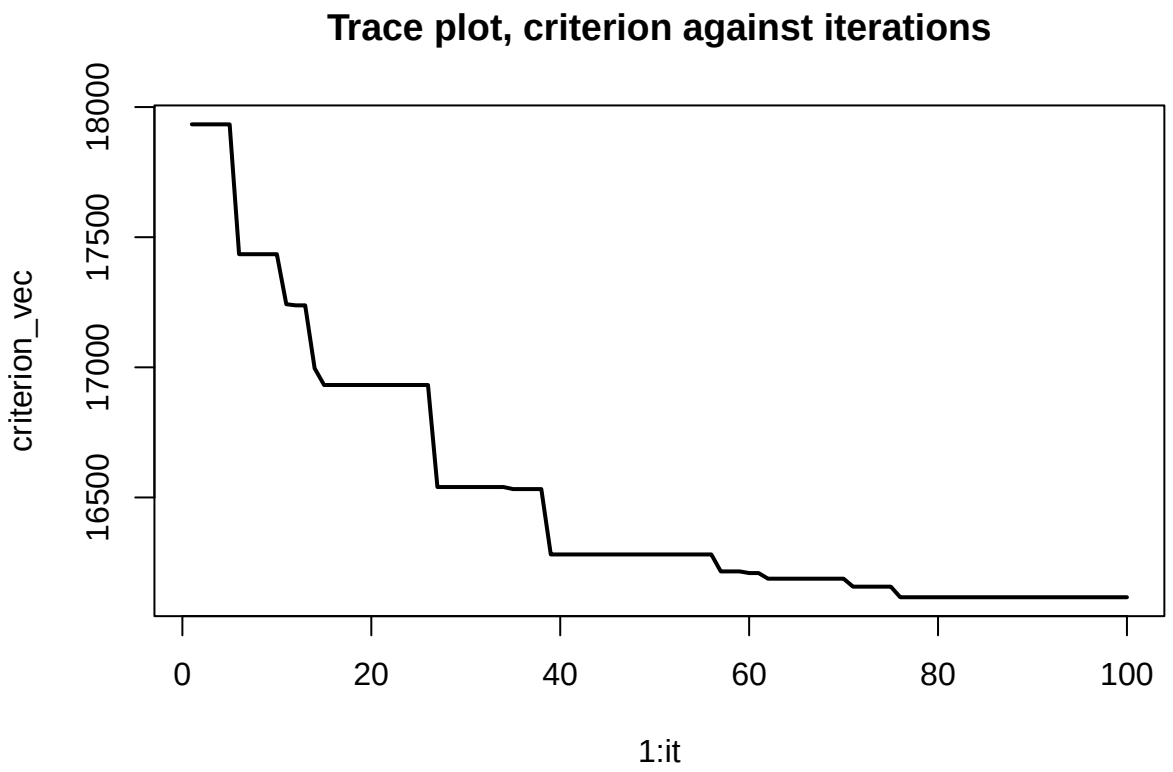
Higher chance of swapping individuals seems to lead to a worse outcome. In the traceplot below we see it is because we do not find improvements as often.

```
## [1] 20
## [1] 40
## [1] 60
## [1] 80
## [1] 100
## [1] "Starting function value:"
## [1] 21936.02
```




```
## [1] "Best function value:"
## [1] 16116.69
## [1] "Average diff:"
## [1] -2120.638
```

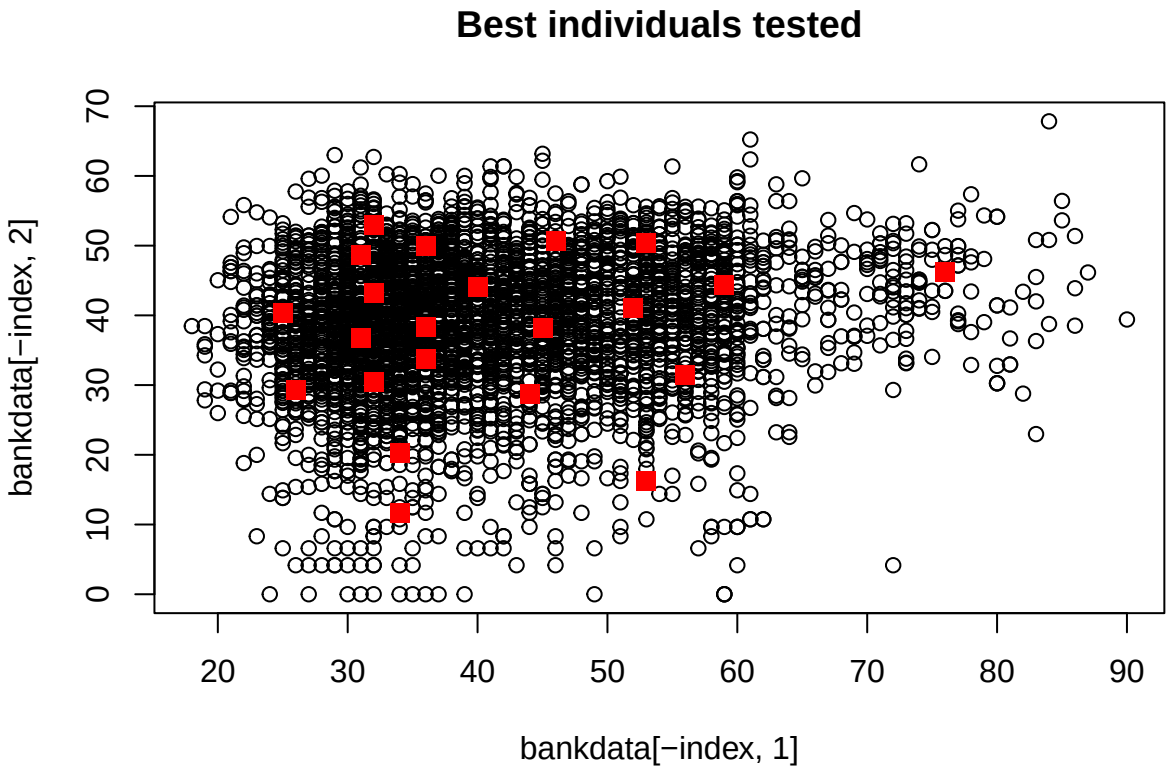

Delta in function
Frequency
0
2000
4000
6000
−10000
−5000
0
5000
diff



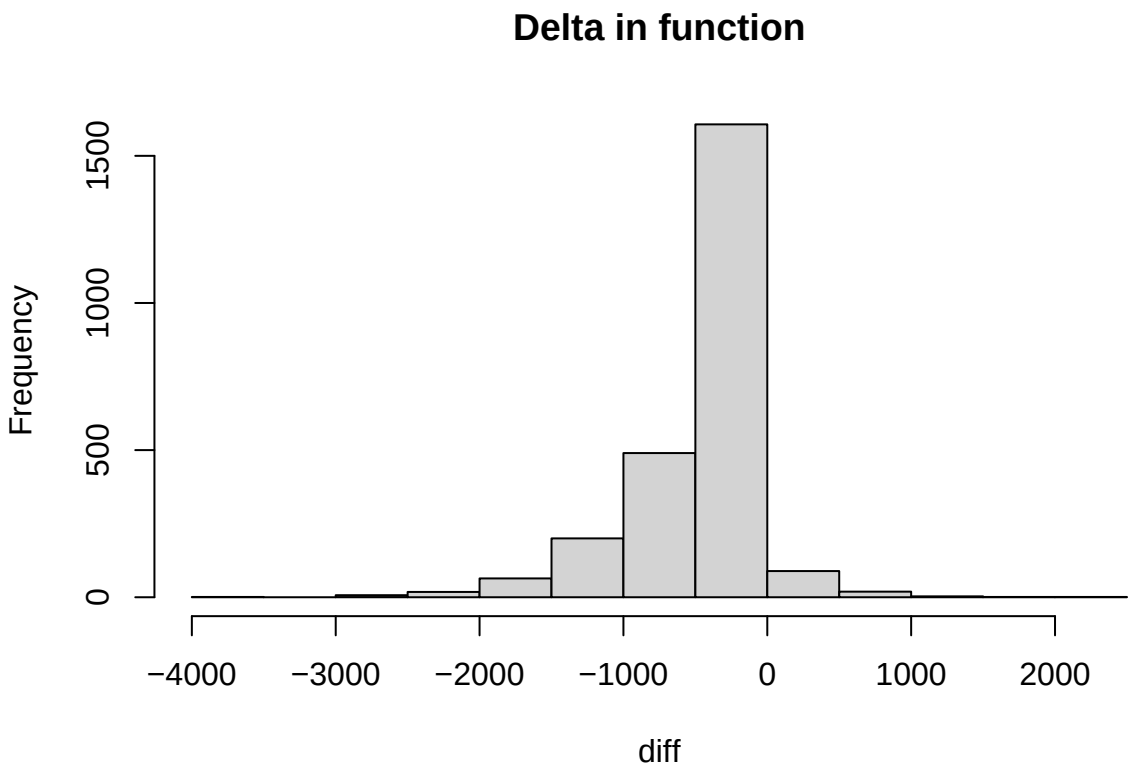

```
##  [1] 3206 1761 1666 2103  755 3976 1021 1223 3352  310 2748 1445 2718 1491 2383
## [16] 3707  348 3043 4115 3184 1036 2309
```

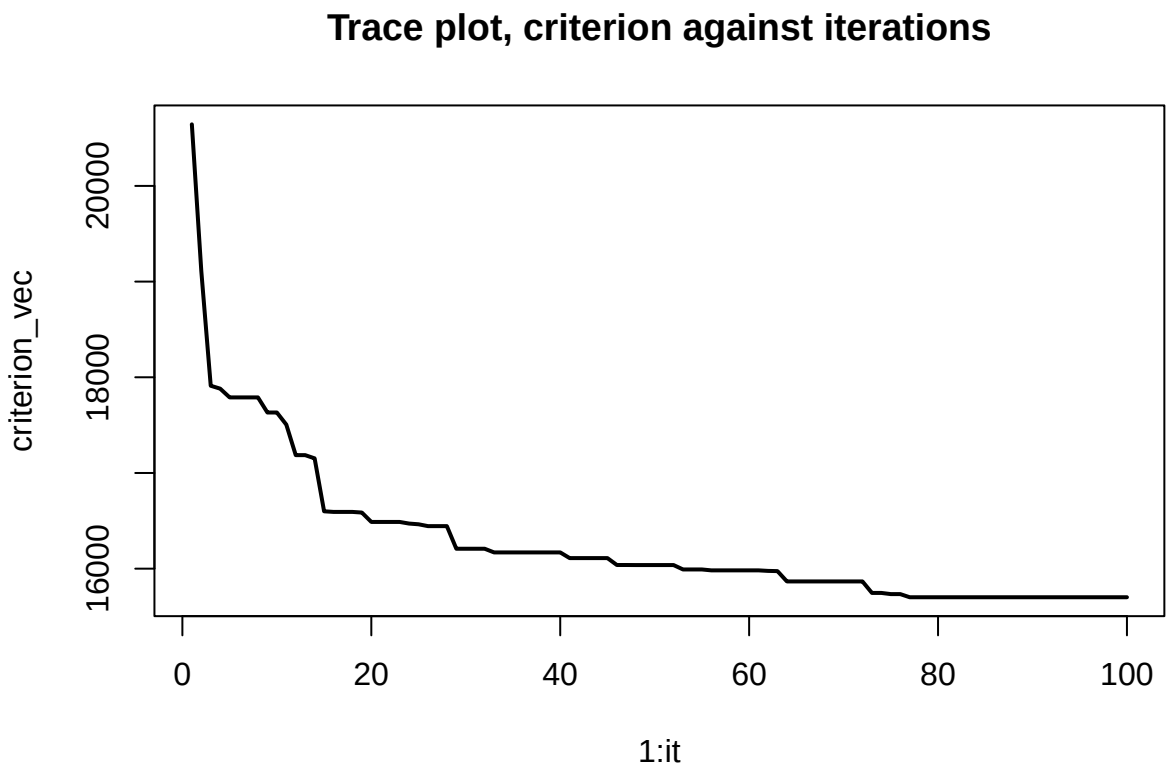
Lower swaprate, on average 1.1 individual performed worse than swapping 2 on average, a fear is that swapping fewer individuals might lead to locked local minima situations or slower progress. In the trace plot we see that we never get the big jumps but like with lower swaprate but we do get steady almost continous progress.

```
## [1] 20
## [1] 40
## [1] 60
## [1] 80
## [1] 100
## [1] "Starting function value:"
## [1] 21936.02
```




```
## [1] "Best function value:"
## [1] 15701.36
## [1] "Average diff:"
## [1] -377.4236
```


Delta in function
Frequency
0
500
1000
1500
−4000
−3000
−2000
−1000
0
1000
2000
diff




```
##  [1] 3795 1786 3510  532 3857 1576 2724 1486   95 4332  367  675  682 3075 3959
## [16]  847 1797  703 2747 2797 1390  916
```

Appendix

```
#################################################################
### Computational statistics, Linköping University, VT2025 ###
### EM algorithm for univariate normal mixtures             ###
### (mixture of two normal distributions)                   ###
### Frank Miller                                            ###
#################################################################

cossim <- function(v1, v2) {
      t(v2)%*%v1 / sqrt(t(v2)%*%v2 * t(v1)%*%v1)


}


emalg <- function(dat, eps=0.000001){

  n      <- length(dat)
  pi_mat <- matrix(NA, nrow=n, ncol=3)
```

```r
# define reasonable starting values for parameters; here based on summary statistics for total datase
p1 <- 1/3
p2 <- 1/3
p3 <- 1/3                # starting value for mixing parameter

sigma1 <- sd(dat)*2/3        # starting value for standard deviation in group taken as 2/3 of total s
sigma2 <- sigma1
sigma3 <- sigma1

mu1    <- mean(dat) + sigma1/2 # starting values for means, taken a bit lower and a bit higher than o
mu2    <- mean(dat) - sigma1/2
mu3 <- mean(dat)
pv     <- c(p1, p2, p3, mu1, mu2, mu3, sigma1, sigma2, sigma3)  # parameter vector

cc     <- eps + 100          # initialize convergence criterion to avoid stopping the while-loop dire

it <- 2

pi_hist <- matrix(1/3, 1, 3)
mu_hist <- matrix(c(mu1, mu2, mu3), 1, 3)
sigma_hist <- matrix(sigma1, 1, 3)

while (cc>eps){
  pv1  <- pv                 # save previous parameter vector
  ### E step ###
  for (j in 1:n){
    w1 <- p1 * dnorm(dat[j], mean=mu1, sd=sigma1)
    w2 <- p2 * dnorm(dat[j], mean=mu2, sd=sigma2)
    w3 <- p3 * dnorm(dat[j], mean=mu3, sd=sigma3)

    sum_w <- w1 + w2 + w3
    pi_mat[j, ] <- c(w1, w2, w3) / sum_w




  }
  ### M step ###
  p1 <- mean(pi_mat[, 1])
  p2 <- mean(pi_mat[, 2])
  p3 <- mean(pi_mat[, 3])

  pi_hist <- rbind(pi_hist, c(p1, p2, p3))



  # Update means
  mu1 <- sum(pi_mat[, 1]* dat)/(p1*n)
  mu2 <- sum(pi_mat[, 2]*dat)/(p2*n)
  mu3 <- sum(pi_mat[, 3]*dat)/(p3*n)
  mu_hist <- rbind(mu_hist, c(mu1, mu2, mu3))
```

```
    # Update standard deviations
    sigma1 <- sqrt(sum(pi_mat[, 1]*(dat - mu1)^2)/(p1*n))
    sigma2 <- sqrt(sum(pi_mat[, 2]*(dat - mu2)^2)/(p2*n))
    sigma3 <- sqrt(sum(pi_mat[, 3]*(dat - mu3)^2)/(p3*n))
    sigma_hist <- rbind(sigma_hist, c(sigma1, sigma2, sigma3))


    ######
    pv <- c(p1, p2, p3, mu1, mu2, mu3, sigma1, sigma2, sigma3)



    cc     <- mean(1-cossim(pv[1:3], pv1[1:3]), 1-cossim(pv[4:6], pv1[4:6]), 1-cossim(pv[7:9], pv1[7:9]))

    it <- it + 1
  }

  return(list(pi_hist = pi_hist, mu_hist = mu_hist, sigma_hist = sigma_hist, pv = pv))
}
# example data and run of the algorithm
data <- c(0.1, 0.5, 0.7, 1.1, 2.5, 3.4, 3.5, 3.9, 4.0)

emalg(data)$pv
emalg(data*1000)$pv

data <- load(file = "threepops.Rdata")

emalg(dat3p)$pv

em <- emalg(dat3p, eps=0.0000000001)

plot_fun <- function(matrix, main) {
  n <- nrow(matrix)
  ylim <- c(min(matrix) * 0.9, max(matrix) * 1.1)

  plot(x = 1:n, y = matrix[, 1], lwd = 2, t = "l", ylim = ylim, main = main)
  lines(x = 1:n,
      y = matrix[, 2],
      col = "blue",
      lwd = 2)
  lines(x = 1:n,
      y = matrix[, 3],
      col = "red",
      lwd = 2)

  legend("topleft", legend = c("Comp 1", "Comp 2", "Comp 3"), col = c("black", "blue", "red"), lwd = 2)
}

plot_fun(em$pi_hist, "pi")
plot_fun(em$mu_hist, "mu")
plot_fun(em$sigma_hist, "sigma")
```

```r
#####################################################################
### Computational statistics, Linköping University, VT2025 ###
### Criterion function Lab 6, Q2 (space filling design)    ###
### Frank Miller                                           ###
###                                                        ###
### We are using in this lab partial data from the         ###
### original bankdata available at                         ###
### https://archive.ics.uci.edu/ml/datasets/Bank+Marketing ###
### See also the publication:                              ###
### Sérgio Moro, P. Cortez, P. Rita (2014). A data-driven  ###
### approach to predict the success of bank telemarketing. ###
### Decision Support Systems.                              ###
#####################################################################

# you need to save the following dataset at the right place and/or add/set the path where it is located
load("bankdata.Rdata")

nclients <- dim(bankdata)[1]  # number of individuals in the dataset, here 4364

# criterion function: sum of minimal distance to an element in the subset
# dat is the full dataset (here: bankdata), subs is the set of ids for the subset selected
# subs should be a vector of elements in 1, ..., 4364; for this question, it should be of length 22
# example call: crit(bankdata, 1:22), selecting the first 22 individuals
# result of this function is the criterion to be minimized
crit <- function(dat, subs){
  s <- length(subs)
  dist <- matrix(rep(NA, nclients*s), ncol=s)
  for (i in 1:s){
    dist[, i] <- sqrt((dat[,1]-dat[subs[i],1])^2+(dat[,2]-dat[subs[i],2])^2)
  }
  sum(apply(dist, 1, min))
}

# it is good to identify the individuals in the full set by their id (1, ..., 4364),
# then we can sample from this set for the starting subset:
fullset <- 1:nclients


set.seed(123)
index <- sample(fullset, 22)

plot(bankdata[-index, 1], bankdata[-index, 2])
points(bankdata[index, 1], bankdata[index, 2], col = "red", cex = 10, pch = 46)



SA <- function(prob_keep = 0.5, alpha = 0.8, beta = 0.5, temp = 4000, mj = 10, it = 100) {
  set.seed(123)
index <- sample(fullset, 22)

diff <- c()

it <- it
```

```r
x_t <- index

prob_keep <- prob_keep

alpha <- alpha
beta <- beta

mj <- mj
temp <- temp

min_function <- crit(bankdata, index)
min_x <- x_t

function_x_t <- min_function

criterion_vec <- c()

for (j in 1:it) {
  if (j %% (round(it/5)) == 0) {
    print(j)
  }
  for (i in 1:mj) {

  swapped_individuals <- prob_keep < runif(22)

  if (sum(swapped_individuals) == 0) {
    x_star <- x_t
  } else {
    not_swapped <- x_t[!swapped_individuals]
    # swapping individuals that got randomly kicked.
    x_star <- c(not_swapped, sample(setdiff(fullset, x_t), sum(swapped_individuals)))
  }

  function_x_star <- crit(bankdata, x_star)
  if (function_x_star < min_function) {
    min_function <- function_x_star
    min_x <- x_star
  }

  delta <- function_x_t - function_x_star
  hx <- exp(delta/temp)
  diff <- c(diff, delta)

  U <- runif(1)
    if (U < hx) {
      x_t <- x_star
      function_x_t <- function_x_star
      #otherwise we do not accept x_t
    }
  }
  criterion_vec <- c(criterion_vec, min_function)
  temp <- alpha * temp
  mj <- max(1, floor(beta * mj))
```

```
}

print("Starting function value:")
print(crit(bankdata, index))

index <- min_x

plot(bankdata[-index, 1], bankdata[-index, 2], main = "Best individuals tested")
points(bankdata[index, 1], bankdata[index, 2], col = "red", cex = 10, pch = 46)

print("Best function value:")
print(min_function)

print("Average diff:")
print(mean(diff))
hist(diff, main = "Delta in function")

plot(x = 1:it, y = criterion_vec, lwd = 2, t = "l", main = "Trace plot, criterion against iterations")

return(min_x)
}



#SA(prob_keep = 0.5, alpha = 0.9, beta = 1.02, mj = 50, temp = 10000, it = 200)
# This gives 16203 as minimum function value but is painfully slow


set.seed(1234)
SA(prob_keep = 0.9, alpha = 0.7, beta = 1.03, mj = 50, temp = 10000, it = 100)


set.seed(1234)
SA(prob_keep = 0.7, alpha = 0.7, beta = 1.03, mj = 50, temp = 10000, it = 100)

set.seed(12345)
SA(prob_keep = 0.95, alpha = 0.7, beta = 1.03, mj = 25, temp = 10000, it = 100)
```